# Class 10: Structural Bioinformatics 1

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## PDB Statistics

The Protein Data Bank (PDB) is the main repository of biomolecular structures. Let's see what it contains:

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```r
url <- "https://tinyurl.com/pdbstats26"
stats <- read.csv("pdb_stats.csv")
head(stats)
```

```
           Molecular.Type   X.ray     EM    NMR Integrative Multiple.methods
1          Protein (only) 178,795 21,825 12,773         343              226
2 Protein/Oligosaccharide  10,363  3,564     34           8               11
3              Protein/NA   9,106  6,335    287          24                7
4     Nucleic acid (only)   3,132    221  1,566           3               15
5                   Other     175     25     33           4                0
6  Oligosaccharide (only)      11      0      6           0                1
  Neutron Other   Total
1      84    32 214,078
2       1     0  13,981
3       0     0  15,759
4       3     1   4,941
5       0     0     237
6       0     4      22
```

The comma is these numbers leads to the numbers here being read as characters. So, let's use a different read:

```r
library(readr)
stats <- read_csv("pdb_stats.csv")
```

```
Rows: 6 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
stats
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
  <chr>                 <dbl> <dbl> <dbl>       <dbl>               <dbl>   <dbl>
1 Protein (only)       178795 21825 12773         343                 226      84
2 Protein/Oligosacch~   10363  3564    34           8                  11       1
3 Protein/NA             9106  6335   287          24                   7       0
4 Nucleic acid (only)    3132   221  1566           3                  15       3
5 Other                   175    25    33           4                   0       0
6 Oligosaccharide (o~      11     0     6           0                   1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

```r
n.xray <- sum(stats$`X-ray`)
n.em <- sum(stats$EM)
n.total <- sum(stats$Total)

n.xray/n.total
```

```
[1] 0.8095077
```

```r
n.em/n.total
```

```
[1] 0.1283843
```

```r
(n.xray + n.em)/n.total
```

```
[1] 0.937892
```

93.7892% are solved by X-Ray, and 5.08% by EM.

> Q2: What proportion of structures in the PDB are protein?

```r
stats[1,9]/n.total
```

```
      Total
1 0.8596889
```

85.96889% of structures are proteins.

> Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

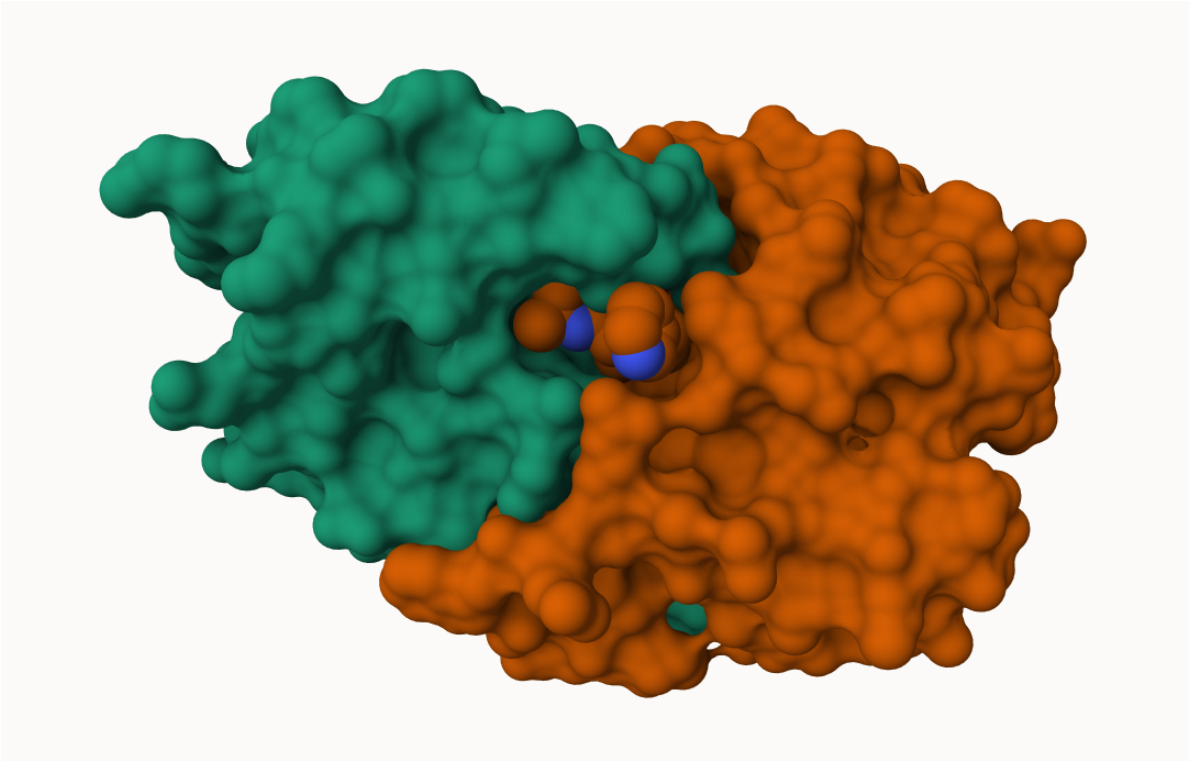
In the current PDB, there are 4,940 structures of HIV-1 protease.

## Visualizing the HIV-1 Protease Structure

We can use the Molstar viewer online: "https://molstar.org/viewer/"

knitr::include_graphics("1HSG.png")

## You can do this by:



A new clean image shwoing the caalytic ASP25 amino acids in both chains of the HIV-Pr dimer along wiht the inhibitor and the all important active site water.

## You can do this by:

knitr::include_graphics("1HSG_2.png")

> Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

We only see one atom per water molecule in this structure because hydrogens are too small to be detected in X-ray crystallography at this resolution.

> Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

The conserved water molecule is HOH 301/302, depending on chain. It is hydrogen-bonded to the inhibitor and ASP 25.

> Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

Discussion Topic: Can you think of a way in which individual, or even larger ligands and substrates, could enter the binding site?

Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

## Bio3D Package for Structural Bioinformatics

**Bio3D** is an R package for structural bioinformatics. Features include the ability to read, write and analyze biomolecular structure, sequence and dynamic trajectory data.

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
pdb
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```r
attributes(pdb)
```

```
$names
[1] "atom"   "xyz"    "seqres" "helix"  "sheet"  "calpha" "remark" "call"

$class
[1] "pdb" "sse"
```

Q7: How many amino acid residues are there in this pdb object?

1514

Q8: Name one of the two non-protein residues?

HOH (127) and MK1 (1)

Q9: How many protein chains are in this structure?

2

### Quick PDB Visualization

Use `library(bio3dview)`, `view.pdb(pdb)`, `sele <- atom.select(pdb, resno=25)`, and `view.pdb(pdb, cols=c("salmon","pink"), highlight = sele, highlight.style = "spacefill") |> setRock()` for visualization.

```
# library(bio3dview)

# view.pdb(pdb)

# Select the important ASP 25 residue
# sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
# view.pdb(pdb, cols=c("salmon","pink"),
      #   highlight = sele,
      #   highlight.style = "spacefill") |>
  # setRock()
```

### Predicting Functional Motions of a Single Structure

Let's read a new PDB structure of Adenylate Kinase and perform Normal mode analysis.

```
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
adk
```

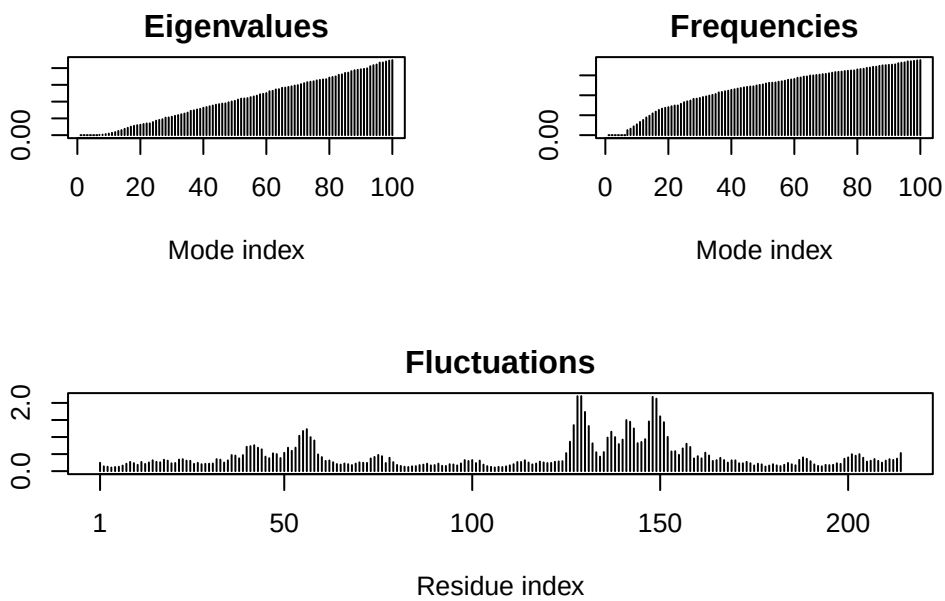
```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
```

```
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
# Perform flexiblity prediction
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.035 seconds.
 Diagonalizing Hessian...   Done in 0.366 seconds.
```

```r
plot(m)
```




Write out results as a trajectory/movie of predicted motions:

```
# mktrj(m, file="adk_m7.pdb") # Viewed on Molt* by uploading "adk_m7.pdb"

# view.nma(m, pdb=adk)
```

### Comparative structure analysis of Adenylate Kinase

The goal of this section is to perform **principal component analysis (PCA)** on the complete collection of Adenylate kinase structures in the protein data-bank (PDB).

First, install by `install.packages("BiocManager")` and `BiocManager::install("msa")`

> Q10. Which of the packages above is found only on BioConductor and not CRAN?

`msa` is only available on BioConductor.

> Q11. Which of the above packages is not found on BioConductor or CRAN?

`bio3dview` is not found on BioConductor or CRAN.

> Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True, `pak` supports installation from GitHub, BitBucket, and other repositories.

```
library(bio3d)
aa <- get.seq("1ake_A")
```

```
Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta


Fetching... Please wait. Done.
```

```
aa
```

```
             1        .         .         .         .         .         60
pdb|1AKE|A   MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
             1        .         .         .         .         .         60

            61        .         .         .         .         .         120
pdb|1AKE|A   DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
            61        .         .         .         .         .         120

           121        .         .         .         .         .         180
```

```
pdb|1AKE|A   VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
           121        .         .         .         .         .         180

           181        .         .         .   214
pdb|1AKE|A   YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
           181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

This sequence is 214 base pairs long.